Reg. No									

St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore – 607001

EPH509S – DIGITAL ELECTRONICS & MICROPROCESSOR

Time : 3 hrs **Max Marks :75**

SECTION – A (10X2=20)

Answer **ALL** Questions

1. Draw the symbol of AND gate and write its truth table.
2. Convert 11001011_2 into octal and hexa decimal equivalents.
3. What is a quad in a Karnaugh map?
4. Draw the circuit diagram for the output Y= A+B+C+D.
5. What is a parity generator?
6. Give the truth table for a Half Adder circuit.
7. Define resolution of a D/A converter.
8. Give the advantages of binary ladder network over variable resistor network.
9. What are the uses of data bus?
10. What do you mean by stack?

SECTION – B (5X5=25)

Answer **ALL** Questions

11. a) Perform the following arithmetic operations in binary number system.
i) 85+52 and ii) 110-32

(or)

b) Explain the construction and working of AND and OR gates using transistors.

12. a) Draw the timing diagram for the NOR-NOR circuit.

(or)

b) Draw the logic circuits for the following Boolean equations:
i) $Y= ABC + AB\overline{C}$ and ii) $\overline{Y} = (\overline{A}+B).(\overline{C}+\overline{D})$

13. a) Explain the working of a JK flip-flop with its truth table.

(or)

b) Draw the circuit diagram of a Full adder circuit and explain its working with its truth table.

14. a) For a 4 bit ladder D/A converter, calculate the output for a digital input of 1010. Also find, full scale voltage when LSB=1 and remaining bits 0. Assume 0=0V and 1=8V.

(or)

b) For an 8 bit counter type A/D converter driven by 300 kHz clock, find out the
i) Maximum conversion time ii) Average conversion time and iii) Maximum conversion rate.

15. a) Write an assembly language program to perform the following task: Load A with 20H and subtract 10H from it. Multiply the result by 2 and store the result in accumulator.

(or)

b) What is an EPROM? What is its advantage over ROM or PROM?

SECTION –C (3X10=30)

Answer any **THREE** Questions

16. Draw the symbol and write the truth table of NOR gate. Show that NOR is universal building block.

17. Explain the simplification of the following expression using K-map.
$Y= F(A,B,C,D) = \sum (0,2,5,7,8,10,13,15)$.

18. Explain the working of
    i) Half subtractor and
    ii) Full subtractor with necessary circuit diagram

19. Explain the principle of simultaneous A/D converter. What is the main advantage and disadvantage of this method?

20. List the various interrupts available in µp 8085 and their priorities.

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